

AGUTS

Ahmedabad Gandhinagar Unified Transit System

BCNF Normalization Proofs

BCNF Proofs (Corrected v2)

Definition: R is in BCNF iff for every non-trivial FD $X \rightarrow Y$ in R, X is a superkey of R.

Key notes for corrected relations:

- INTERCHANGE and SMART_CARD each have two candidate keys. For BCNF, both determinants must be superkeys.
- ROUTE_STOP now has a 3-part composite PK: {route_id, stop_id, sequence_no}.
- STAFF_CERTIFICATION and PASSENGER_PHONE are new multi-valued attribute relations.

Summary — All 27 Relations

#	Relation	Primary Key	Non-Trivial FDs	BCNF?
1	CITY	{city_id}	city_id $\rightarrow$ {city_name, city_state}	✓ Yes
2	TRANSIT_OPERATOR	{operator_id}	operator_id $\rightarrow$ {operator_name, service_type, city_id}	✓ Yes
3	METRO_LINE	{line_id}	line_id $\rightarrow$ {line_name, line_code, operator_id}	✓ Yes
4	FARE_ZONE★	{zone_id}	zone_id $\rightarrow$ {zone_name, operator_id}	✓ Yes
5	DEPOT★	{depot_id}	depot_id $\rightarrow$ {depot_name, address, operator_id, city_id}	✓ Yes
6	VEHICLE	{vehicle_id}	vehicle_id $\rightarrow$ {reg_no, capacity, mfr_year, mfr_name, mfr_info, status, operator_id, depot_id}	✓ Yes
7	BUS	{vehicle_id}	vehicle_id $\rightarrow$ {fuel_type, bus_type}	✓ Yes
8	METRO_TRAIN★	{vehicle_id}	vehicle_id $\rightarrow$ {rake_number, num_coaches}	✓ Yes
9	MAINTENANCE★	{maintenance_id}	maintenance_id $\rightarrow$ {vehicle_id, start_date, end_date, maint_type, description}	✓ Yes
10	STOP★	{stop_id}	stop_id $\rightarrow$ {stop_name, stop_type, latitude, longitude, city_id}	✓ Yes
11	INTERCHANGE★	{interchange_id, stop_id (2nd CK)}	interchange_id $\rightarrow$ {stop_id, type, desc} AND stop_id $\rightarrow$ {iid, type, desc}	✓ Yes

1 2	ROUTE★	{route_id}	route_id → {route_no, route_name, dist_km, operator_id, line_id}	✓ Yes
1 3	ROUTE_STOP★	{route_id, stop_id, seq_no}	{ route_id, stop_id, seq_no } → {zone_id}	✓ Yes
1 4	STAFF★	{staff_id}	staff_id → {first_name, middle_name, last_name, role, contact, doj, operator_id, depot_id}	✓ Yes
1 5	STAFF_CERTIFICATION★	{staff_id, certification}	{staff_id, certification} → {issued_date, expiry_date}	✓ Yes
1 6	DRIVER	{staff_id}	staff_id → {license_no}	✓ Yes
1 7	CONDUCTOR	{staff_id}	staff_id → {badge_no}	✓ Yes
1 8	MOTORMAN	{staff_id}	staff_id → {metro_license_no}	✓ Yes
1 9	TRIP★	{trip_id}	trip_id → {trip_date, trip_status, route_id, vehicle_id, staff_id}	✓ Yes
2 0	TRIP_STOP★	{trip_id, sequence_no}	{trip_id, sequence_no} → {stop_id, sched_arrival, actual_arrival}	✓ Yes
2 1	PASSENGER★	{passenger_id}	passenger_id → {first_name, middle_name, last_name, email, dob, concession, city_id}	✓ Yes
2 2	PASSENGER_PHONE★	{passenger_id, phone}	{passenger_id, phone} → {phone_type}	✓ Yes
2 3	SMART_CARD	{card_id, passenger_id (2nd CK)}	card_id → {passenger_id, balance, issue_date, expiry_date, card_status}	✓ Yes
2 4	PASS★	{pass_id}	pass_id → {pass_type, valid_from, valid_to, amount_paid, passenger_id, operator_id}	✓ Yes
2 5	FARE★	{fare_id}	fare_id → {fare_amount, concession_type, operator_id, from_zone_id, to_zone_id}	✓ Yes
2 6	TICKET_TRANSACTION	{txn_id}	txn_id → {ticket_type, amount, txn_datetime, payment_mode, passenger_id, trip_id, card_id}	✓ Yes

2 7	COMPLAINT	{complaint_id}	complaint_id → {category, description, status, filed_date, resolved_date, passenger_id, trip_id}	✓ Yes
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Detailed Proofs — Each Relation

Proof 1: CITY

```
U = {city_id, city_name, city_state}
```

Non-trivial FDs:

```
city_id → city_name  
city_id → city_state
```

Closure computation:

```
{city_id}+ = {city_id, city_name, city_state} = U
```

city_id is a superkey. ✓ **BCNF**

Proof 2: TRANSIT_OPERATOR

```
U = {operator_id, operator_name, service_type, city_id}
```

Non-trivial FDs:

```
operator_id → operator_name  
operator_id → service_type  
operator_id → city_id
```

Closure computation:

```
{operator_id}+ = U
```

operator_id is a superkey. ✓ **BCNF**

Proof 3: METRO_LINE

```
U = {line_id, line_name, line_code, operator_id}
```

Non-trivial FDs:

```
line_id → line_name  
line_id → line_code  
line_id → operator_id
```

Closure computation:

```
{line_id}+ = U
```

line_id is a superkey. ✓ **BCNF**

Proof 4: FARE_ZONE ★

```
U = {zone_id, zone_name, operator_id}
```

Non-trivial FDs:

```
zone_id → zone_name
zone_id → operator_id
```

Closure computation:

```
{zone_id}+ = {zone_id, zone_name, operator_id} = U
```

zone_id is a superkey. operator_id was added as FK; still determined by zone_id. ✓ **BCNF** ★

Proof 5: DEPOT ★

```
U = {depot_id, depot_name, address, operator_id, city_id}
```

Non-trivial FDs:

```
depot_id → depot_name
depot_id → address
depot_id → operator_id
depot_id → city_id
```

Closure computation:

```
{depot_id}+ = {depot_id, depot_name, address, operator_id, city_id} = U
```

depot_id is a superkey. city_id FK added; still determined solely by depot_id. ✓ **BCNF** ★

Proof 6: VEHICLE

```
U = {vehicle_id, registration_no, capacity, manufacture_year,
manufacturer_name, manufacturer_info, status, operator_id, depot_id}
```

Non-trivial FDs:

```
vehicle_id → registration_no
vehicle_id → capacity
... → all others
```

Closure computation:

```
{vehicle_id}+ = U (applying all 8 FDs)
```

vehicle_id is a superkey. ✓ **BCNF**

Proof 7: BUS

```
U = {vehicle_id, fuel_type, bus_type}
```

Non-trivial FDs:

```
vehicle_id → fuel_type
```

```
vehicle_id → bus_type
```

Closure computation:

```
{vehicle_id}+ = U
```

vehicle_id is a superkey (ISA PK from VEHICLE). ✓ **BCNF**

Proof 8: METRO_TRAIN ★

```
U = {vehicle_id, rake_number, num_coaches}
```

Non-trivial FDs:

```
vehicle_id → rake_number
```

```
vehicle_id → num_coaches
```

Closure computation:

```
{vehicle_id}+ = U
```

vehicle_id is a superkey. Renamed train_number → rake_number; BCNF unchanged. ✓
BCNF ★

Proof 9: MAINTENANCE ★

```
U = {maintenance_id, vehicle_id, start_date, end_date, maintenance_type,  
description}
```

Non-trivial FDs:

```
maintenance_id → vehicle_id
```

```
maintenance_id → start_date
```

```
maintenance_id → end_date
```

```
maintenance_id → maintenance_type
```

```
maintenance_id → description
```

Closure computation:

```
{maintenance_id}+ = U
```

maintenance_id is a superkey. New column maintenance_type still determined by maintenance_id. ✓ **BCNF** ★

Proof 10: STOP ★

```
U = {stop_id, stop_name, stop_type, latitude, longitude, city_id}
```

Non-trivial FDs:

```
stop_id → stop_name  
stop_id → stop_type  
stop_id → latitude  
stop_id → longitude  
stop_id → city_id
```

Closure computation:

```
{stop_id}+ = U
```

stop_id is a superkey. zone_id removed from this table; no impact on BCNF. ✓ **BCNF** ★

Proof 11: INTERCHANGE ★

```
U = {interchange_id, stop_id, interchange_type, description}
```

Non-trivial FDs:

```
interchange_id → stop_id    (F1)  
interchange_id → interchange_type  (F2)  
interchange_id → description  (F3)  
stop_id → interchange_id    (F4, because UNIQUE constraint on stop_id)
```

Closure computation:

```
Two candidate keys: {interchange_id} and {stop_id}.
```

```
{interchange_id}+: apply F1,F2,F3 → {interchange_id, stop_id,  
interchange_type, description} = U
```

```
{stop_id}+: apply F4 → interchange_id; apply F2,F3 via interchange_id →  
{stop_id, interchange_id, interchange_type, description} = U
```


Both candidate keys are superkeys. Every non-trivial FD has a superkey as its LHS. ✓ **BCNF** ★

Proof 12: ROUTE ★

```
U = {route_id, route_number, route_name, distance_km, operator_id, line_id}
```

Non-trivial FDs:

```
route_id → route_number
route_id → route_name
route_id → distance_km
route_id → operator_id
route_id → line_id
```

Closure computation:

```
{route_id}+ = U
```

route_id is a superkey. operator_id and line_id FKs added; both determined by route_id. ✓ **BCNF** ★

Proof 13: ROUTE_STOP ★

```
U = {route_id, stop_id, sequence_no, zone_id}
```

Non-trivial FDs:

```
{route_id, stop_id, sequence_no} → zone_id
```

Closure computation:

```
{route_id, stop_id, sequence_no}+ = {route_id, stop_id, sequence_no,
zone_id} = U
```

{route_id, stop_id, sequence_no} is the 3-part composite PK and a superkey. ✓ **BCNF** ★

Proof 14: STAFF ★

```
U = {staff_id, first_name, middle_name, last_name, role, contact,
date_of_joining, operator_id, depot_id}
```

Non-trivial FDs:

```
staff_id → first_name
staff_id → middle_name
staff_id → last_name
```

```
staff_id → role
staff_id → contact
staff_id → date_of_joining
staff_id → operator_id
staff_id → depot_id
```

Closure computation:

```
{staff_id}+ = U
```

staff_id is a superkey. Name split into sub-attrs; BCNF unchanged. ✓ **BCNF** ★

Proof 15: STAFF_CERTIFICATION ★ (NEW)

```
U = {staff_id, certification, issued_date, expiry_date}
```

Non-trivial FDs:

```
{staff_id, certification} → issued_date
{staff_id, certification} → expiry_date
```

Closure computation:

```
{staff_id, certification}+ = {staff_id, certification, issued_date,
expiry_date} = U
```

{staff_id, certification} is the composite PK and a superkey. New relation is in BCNF. ✓ **BCNF** ★

Proof 16: DRIVER

```
U = {staff_id, license_no}
```

Non-trivial FDs:

```
staff_id → license_no
```

Closure computation:

```
{staff_id}+ = U
```

staff_id is a superkey. ✓ **BCNF**

Proof 17: CONDUCTOR

```
U = {staff_id, badge_no}
```

Non-trivial FDs:

```
staff_id → badge_no
```

Closure computation:

```
{staff_id}+ = U
```

staff_id is a superkey. ✓ **BCNF**

Proof 18: MOTORMAN

```
U = {staff_id, metro_license_no}
```

Non-trivial FDs:

```
staff_id → metro_license_no
```

Closure computation:

```
{staff_id}+ = U
```

staff_id is a superkey. ✓ **BCNF**

Proof 19: TRIP ★

```
U = {trip_id, trip_date, trip_status, route_id, vehicle_id, staff_id}
```

Non-trivial FDs:

```
trip_id → trip_date
```

```
trip_id → trip_status
```

```
trip_id → route_id
```

```
trip_id → vehicle_id
```

```
trip_id → staff_id
```

Closure computation:

```
{trip_id}+ = U
```

trip_id is a superkey. Derived attrs (etd, eta, delay_minutes) correctly removed. ✓ **BCNF** ★

Proof 20: TRIP_STOP ★

```
U = {trip_id, sequence_no, stop_id, sched_arrival, actual_arrival}
```

Non-trivial FDs:

```
{trip_id, sequence_no} → stop_id
```

```
{trip_id, sequence_no} → sched_arrival
```

```
{trip_id, sequence_no} → actual_arrival
```

Closure computation:

```
{trip_id, sequence_no}+ = U
```

Composite PK is a superkey. stop_order renamed to sequence_no. ✓ BCNF ★

Proof 21: PASSENGER ★

```
U = {passenger_id, first_name, middle_name, last_name, email, date_of_birth,
      concession_type, city_id}
```

Non-trivial FDs:

```
passenger_id → first_name
passenger_id → middle_name
passenger_id → last_name
passenger_id → email
passenger_id → date_of_birth
passenger_id → concession_type
passenger_id → city_id
```

Closure computation:

```
{passenger_id}+ = U
```

passenger_id is a superkey. Phone removed; name split; BCNF unchanged. ✓ BCNF ★

Proof 22: PASSENGER_PHONE ★ (NEW)

```
U = {passenger_id, phone, phone_type}
```

Non-trivial FDs:

```
{passenger_id, phone} → phone_type
```

Closure computation:

```
{passenger_id, phone}+ = {passenger_id, phone, phone_type} = U
```

{passenger_id, phone} is the composite PK and a superkey. New MV-attribute relation is in BCNF. ✓ BCNF ★

Proof 23: SMART_CARD

```
U = {card_id, passenger_id, balance, issue_date, expiry_date, card_status}
```

Non-trivial FDs:

```
card_id → passenger_id  (F1)
card_id → balance      (F2)
card_id → issue_date   (F3)
card_id → expiry_date  (F4)
card_id → card_status  (F5)
passenger_id → card_id  (F6, UNIQUE constraint)
```

Closure computation:

```
Two candidate keys: {card_id} and {passenger_id}.
{card_id}+: apply F1-F5 → U
{passenger_id}+: apply F6 → card_id; then F1-F5 → U
Every FD LHS is one of these two superkeys.
```

Both determinants are superkeys. ✓ **BCNF**

Proof 24: PASS ★

```
U = {pass_id, pass_type, valid_from, valid_to, amount_paid, passenger_id,
operator_id}
```

Non-trivial FDs:

```
pass_id → pass_type
pass_id → valid_from
pass_id → valid_to
pass_id → amount_paid
pass_id → passenger_id
pass_id → operator_id
```

Closure computation:

```
{pass_id}+ = U
```

pass_id is a superkey. fare_id replaced by operator_id; BCNF unchanged. ✓ **BCNF** ★

Proof 25: FARE ★

```
U = {fare_id, fare_amount, concession_type, operator_id, from_zone_id, to_zone_id}
```

Non-trivial FDs:

```
fare_id → fare_amount
fare_id → concession_type
fare_id → operator_id
fare_id → from_zone_id
fare_id → to_zone_id
```

Closure computation:

```
{fare_id}+ = U
```

fare_id is a superkey. operator_id FK added; BCNF unchanged. ✓ **BCNF** ★

Proof 26: TICKET_TRANSACTION

```
U = {txn_id, ticket_type, amount, txn_datetime, payment_mode, passenger_id, trip_id, card_id}
```

Non-trivial FDs:

```
txn_id → all others
```

Closure computation:

```
{txn_id}+ = U
```

txn_id is a superkey. ✓ **BCNF**

Proof 27: COMPLAINT

```
U = {complaint_id, category, description, status, filed_date, resolved_date, passenger_id, trip_id}
```

Non-trivial FDs:

```
complaint_id → all others
```

Closure computation:

```
{complaint_id}+ = U
```

complaint_id is a superkey. ✓ **BCNF**